

Titanic Dataset Analysis – Visualization & Design Ethics

Submitted By	Abhiney Kumar
Enrollment No	198409
Email	abhineyk.ec.24@nitj.ac.in
Assignment	Assignment 2 — Summer Internship AI & ML 2026
Submitted To	Dr. Ashok Gopalakrishnan
Date	17 June 2026
Dataset	Titanic Passenger Survival Dataset



Q 1

Which chart best shows Titanic survival by gender and class?
Encourages chart choice connected to data storytelling

When I first looked at the dataset, I noticed I had two categorical variables — passenger class (1st, 2nd, 3rd) and gender — plus the outcome: survived or not. My goal was to show all of this together without cluttering the chart. That made me think — what type of chart lets me compare multiple groups side by side while keeping the numbers visible and the story clear?

My answer: a **Grouped Bar Chart**.

Why not other options?

Chart Type	Why it does not work well here
Pie Chart	Becomes unreadable with 6+ slices. No easy comparison across groups.
Line Graph	Implies a continuous trend. Class is discrete.
Single Bar	Cannot show two variables (gender + class) simultaneously.
Grouped Bar (chosen)	Shows all 6 groups side by side. Easy to compare survived vs did not across class and gender in one glance.

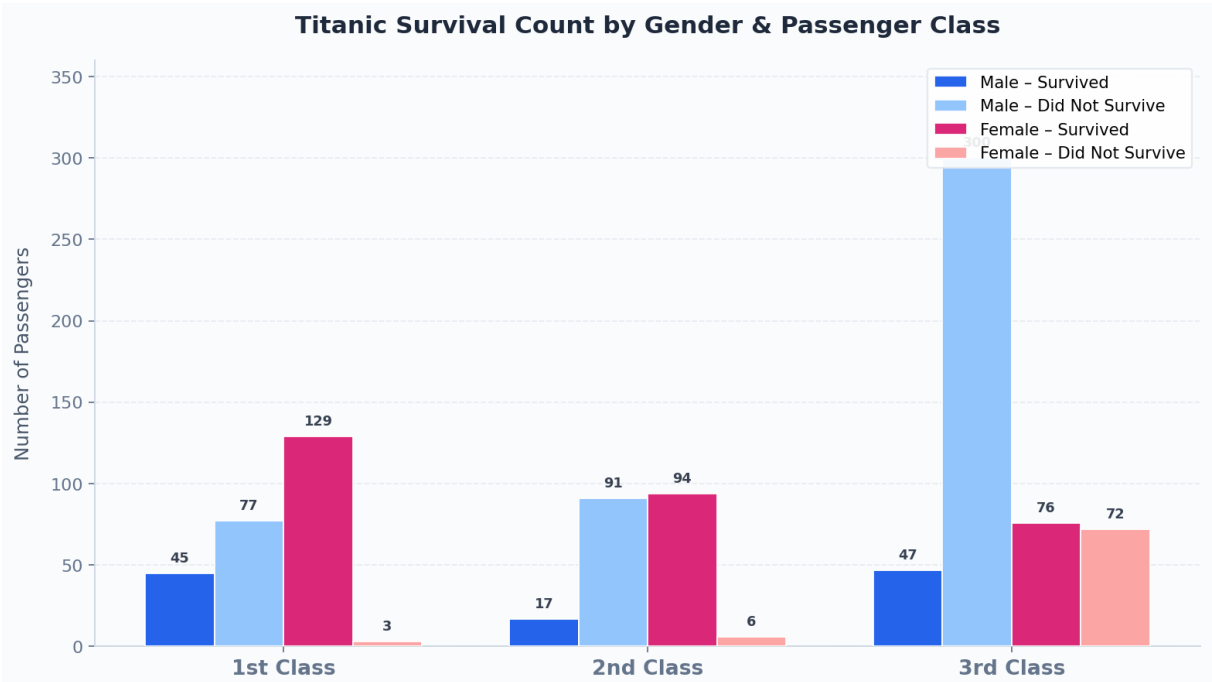


Chart 1 — Grouped Bar Chart: Actual survival count split by gender across 1st, 2nd and 3rd class. Each cluster of 4 bars represents one class. Blue = Male, Pink = Female.

Key Observations from Chart 1:

- 1st Class Females: 129 survived out of 132 — a 97.7% survival rate. The highest of any group.
- 3rd Class Males: Only 47 survived out of 347 — a 13.5% survival rate. The lowest of any group.
- Every class followed the same pattern: females survived at a significantly higher rate than males.
- Class clearly mattered — higher class meant more lifeboats, better deck access, and faster evacuation.

The 100% Stacked Bar — A Complementary View

While the grouped bar shows raw counts, I also wanted to show survival rates in **percentage** — because the groups have different sizes. A 100% stacked bar chart normalises this. Each bar becomes 100% and is split between survived and did not survive, making it easy to compare proportions without the absolute numbers affecting perception.

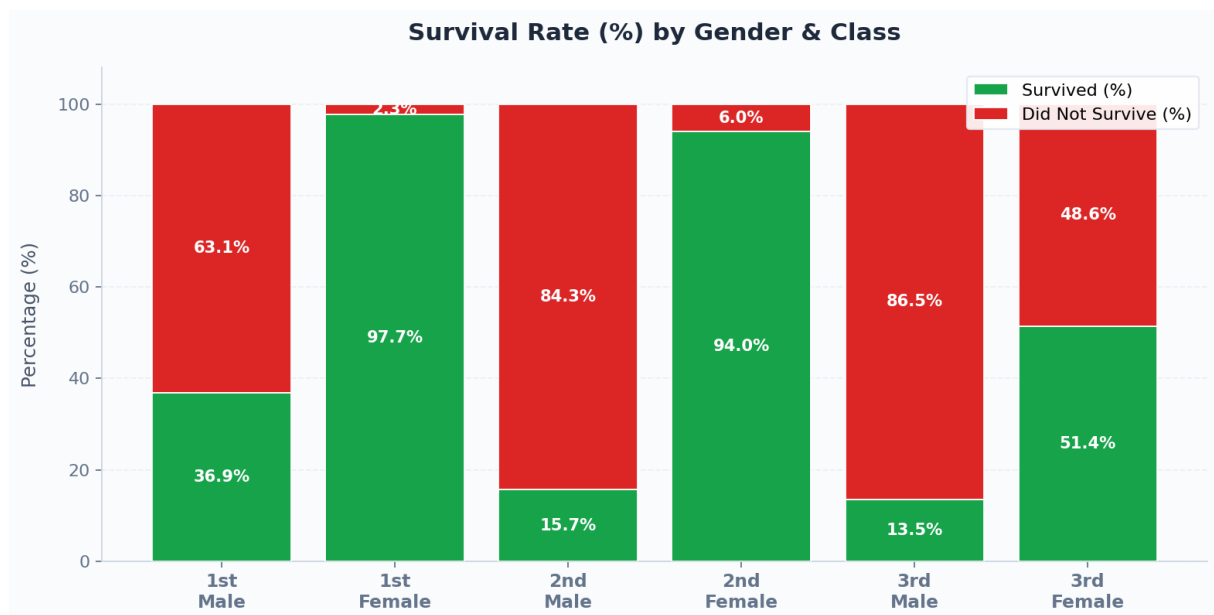


Chart 2 — 100% Stacked Bar: Survival rate (%) per group. 1st class females had 97.7% survival. 3rd class males had only 13.5%.

Together, Charts 1 and 2 tell a complete story — one shows the scale of loss in numbers, the other shows the inequality in chances of survival. That is what good visualization does: it does not just display data, it **reveals the human story inside the numbers**.

Q 2

How can poor design distort the message of a visualization?

Promotes critical thinking about ethics and clarity in data communication

While working on the charts for Question 1, I realised something — the same data can look completely different depending on *how* you present it. A chart is not just a picture of numbers. It is an argument. And just like a misleading argument can distort truth with words, a poorly designed chart can distort truth with visuals.

1. Misleading Scales (Truncated Y-Axis)

This is the most common — and most dangerous — design mistake. When the Y-axis does not start at zero, even a tiny difference between two values looks enormous. I demonstrated this below using two charts of the exact same data:

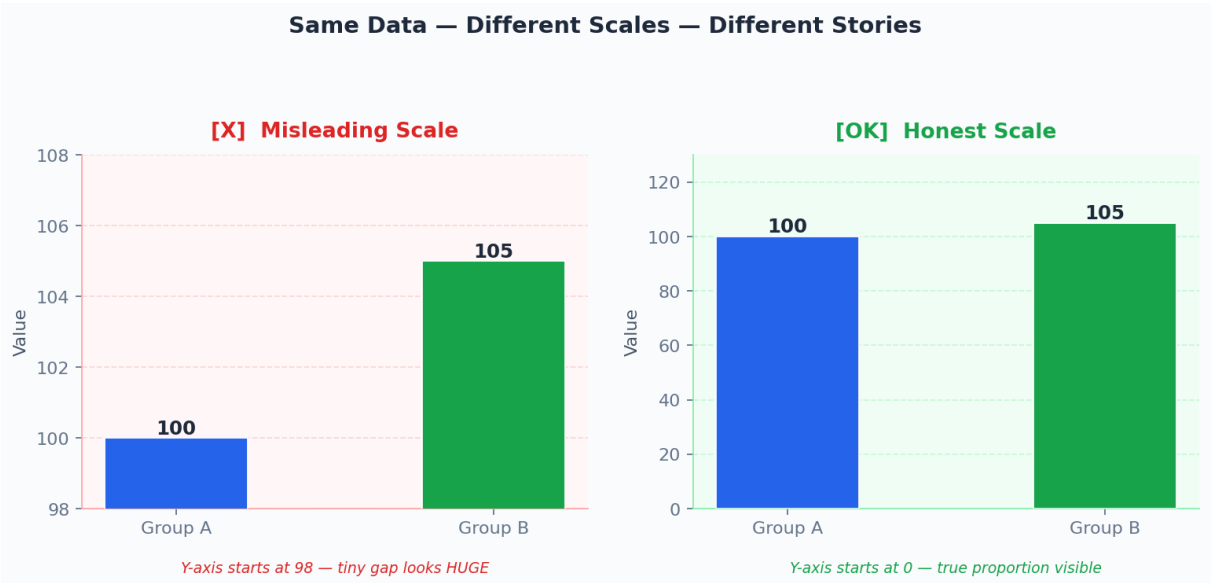


Chart 3 — Same data (Group A = 100, Group B = 105), two different Y-axis scales. Left: axis starts at 98, making a 5% gap look massive. Right: axis starts at 0, true proportion visible.

In real-world scenarios, this trick is used all the time — in news channels, company earnings slides, and political campaign graphics. A stock that dropped from █500 to █490 can be made to look like a 'massive crash' simply by zooming the Y-axis in. The data is technically correct, but the impression it creates is completely false.

2. Excessive or Meaningless Color

Too many colors:	When every bar, every slice, every line gets a different color, the viewer's brain gets overloaded. Color should carry meaning — like red = loss, green = gain — not just be decorative. A rainbow of colors draws attention away from the actual data.
Emotionally wrong colors:	Using red to represent something positive, or green for something negative, goes against how humans instinctively read visual signals. The viewer will feel alarmed even if the data says something good, purely because of color psychology.
3D effects with color shading:	3D pie charts are particularly misleading. The slice nearest to the viewer looks larger due to perspective and shadow — even though the actual value has not changed. Many people have made wrong decisions based on visual tricks like this.

3. Cherry-Picking the Time Range

If a company's revenue fell from 100 to 50 over 5 years but recovered to 80 in the last 6 months, showing only those last 6 months makes it look like a success story. The chart is not lying about the numbers shown — but it is hiding what happened before. This selective framing is a design choice, not a data choice, and it distorts the complete truth.

4. Using the Wrong Chart Type

Even the type of chart changes how data feels. A **line chart implies continuity and trend**. A **bar chart implies comparison**. If I had shown the Titanic survival across classes using a line chart — connecting 1st to 2nd to 3rd

My Takeaway — The Ethics of Visualization:

People trust charts more than they trust paragraphs. When we see a graph, we assume it is an objective truth. But every chart has design choices baked in — what to show, what to hide, which scale to use, which colors to pick, which time range to frame. A good data analyst does not just make charts. They make **honest charts**. The moment you start designing a chart to make data look a certain way rather than show what it actually says, you cross from analysis into manipulation. That is the line we must never cross.

Assignment Summary

The Titanic dataset reveals that gender and class were the two most critical factors in determining survival. A Grouped Bar Chart, supported by a 100% Stacked Bar, communicates this story most clearly. Effective visualization requires honest design — correct scale, purposeful color, appropriate chart type, and complete data framing. Any deviation from these principles, even unintentional, misleads the audience and undermines the entire purpose of data communication.

— Abhiney Kumar | Enrollment No: 198409 | abhineyk.ec.24@nitj.ac.in
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